

Yagnesh Tripathi

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SUMMARY

AI Leader with extensive experience in Data Science and AI, specializing in Generative AI, Multi-Agent AI systems, and LLM-powered solutions. Proven success in designing enterprise-scale AI applications, leading high-performing teams, and deploying production-ready AI/ML solutions on Cloud and on-premise. Strong track record of driving measurable business impact across telecom, insurance, healthcare, and customer experience analytics.

SKILLS

Generative AI & Large Language Models: LLaMa, Qwen, Mistral, RAG (Retrieval-Augmented Generation), Prompt Engineering, Instruction-Tuned Models, Fine-Tuning (LoRA, PEFT, Adapters), Contextual Embeddings

Multi-Agent Systems & Orchestration: Agentic AI, Multi-Agent Systems, LangGraph, LangChain, Memory-Augmented Agents, Agent Collaboration

Vector Databases & Knowledge Retrieval: FAISS, Pinecone, Vector Search, Long-Term Memory, Embedding Models (Sentence-BERT, Alibaba embeddings)

Data Science & Machine Learning: Predictive Modeling, NLP, Deep Learning, Neural Networks, Feature Engineering, Classification, Regression, Clustering, XGBoost, Random Forest, LSTM, Model Evaluation

Cloud & Deployment (MLOps): AWS Bedrock, AWS SageMaker, AWS Lambda, Amazon S3, GCP, Vertex AI, Google Cloud Run, Docker

Programming & Tools: Python, SQL, GitHub, Hugging Face, Transformers, Pytorch

Leadership & Business Skills: AI Strategy, Project Delivery, People Management, Stakeholder Management, Client Engagement

EXPERIENCE

Tech Lead – Data Science

Sutherland Global Services Private Limited

May 2022 – Present, Hyderabad

- Designed and deployed GenAI and multi-agent AI solutions for insurance and telecom industries.
- Led a team of 7 Data Scientists, achieving measurable cost savings and efficiency improvements.
- Delivered AI-driven QA automation, Contact Drivers, and objection-handling simulators.
- Conducted technical interviews, hiring evaluations, and team upskilling sessions.
- Championed the integration of advanced machine learning frameworks and cloud pipelines, bolstering deployment scalability and facilitating rapid model iteration cycles for core client projects.

Continuous Improvement Specialist

Syri Research Pvt. Ltd

November 2017 – December 2021, Vadodara

- Partnered with business teams to deliver analytics-driven strategies and process improvements.
- Built predictive models (regression, classification, clustering) to improve decision-making.
- Automated data analysis pipelines, boosting efficiency and reducing manual work.

Assistant Manager

Optum Global Solutions

June 2016 – May 2017, Hyderabad

- Identified challenges and process improvement opportunities suitable for Six Sigma (DMAIC) methodology, applying a data-driven, statistical approach to problem-solving.
- Actively identified process gaps and coordinated with stakeholders for root cause analysis (RCA), implementation of corrective and preventive actions (CAPA), and successful closure of improvement loops.
- Built dashboards & automated reports to support U.S. healthcare operations.

Process Excellence Specialist

Advantmed LLC

November 2008 – May 2016, Ahmedabad

- Managed a team of 30+ members while working in operations for over five years, ensuring achievement of team targets, service level agreements (SLAs), and overall deliverables.
- Led continuous improvement & process automation initiatives across medical coding operations.

PROJECT

Multi-Agent AI System for Insurance Sales Simulation Coaching

- Designed the system to serve as a training simulator to coach Licensed sales agents, enabling continuous learning from past successes and improving objection-handling strategies through AI-generated alternative responses
- Built a multi-agent AI system using LangGraph to orchestrate autonomous customer and sales agents in realistic insurance failed-sales call scenarios.
- Integrated a locally hosted Mistral 7B CS model for objection classification, LLaMA 3 8B model for context-aware response generation, and handling multiple objections per call
- Implemented long-term memory with FAISS vector store for objection-response retrieval and short-term memory to maintain full conversational context
- Extracted and labeled objection-response pairs from past transcripts to seed memory and continuously refine agent performance.

Call Summarization and Feature Extraction

- Developed a GenAI-based system to summarize customer-agent call transcripts, capturing structured information such as customer intent, objections, agent response, Medical Conditions, and compliance markers.
- Integrated AWS Bedrock for managed LLM summarization and feature extraction, reducing operational overhead of model hosting.
- Stored transcripts and extracted metadata in Amazon S3, enabling scalable and secure access across analytics workflows.
- Automated preprocessing and post-processing with AWS Lambda, ensuring lightweight, serverless orchestration.
- Delivered outputs into CX360 workflows, reducing manual QA effort and accelerating decision-making efficiency.

Natural Language to SQL Model for CX360 Portal

- Developed and deployed Text-to-SQL solutions using Qwen2.5-Coder-14B-Instruct-GPTQ-Int8 model to convert natural language queries into executable SQL code.
- Enabled business users to retrieve insights from databases without writing complex queries, significantly improving data accessibility and self-service analytics on agent performance.
- Collaborated with data engineering and analytics teams to evaluate SQL accuracy, optimize performance, and maintain data governance. Incorporated safety checks and fallback mechanisms to mitigate incorrect or potentially risky query execution in production environments.

Follow Up Call Prioritization For Lead Conversion Prediction

- Developed a machine learning model to predict the likelihood of lead conversion during scheduled follow-up calls, where agents contacted prospects at their preferred date and time after initial engagement.
- Optimized agent effort, personalize outreach strategy, and improve overall conversion rates.
- Achieved measurable improvements in conversion rates from 15% to 18% and lead lifecycle efficiency, while reducing agent idle time and drop-offs in the follow-up funnel.

EDUCATION

Bachelor of Commerce

Gujarat University • Gujarat

CERTIFICATIONS

Six Sigma Black Belt

Indian Statistical Institute, Mumbai • 2016

Six Sigma Green Belt

Indian Statistical Institute, Mumbai • 2015